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Norbert Wiener's Cybernetics

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VI - Gestalt and Universals

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VIII - Languages, Information, Society

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VI - Gestalt and Universals

Norbert Wiener's Cybernetics

Norbert Wiener introduces two concepts that have profound implications for understanding the nature of information processing and communication in this chapter: **Gestalt** and **Universal Recoding.**

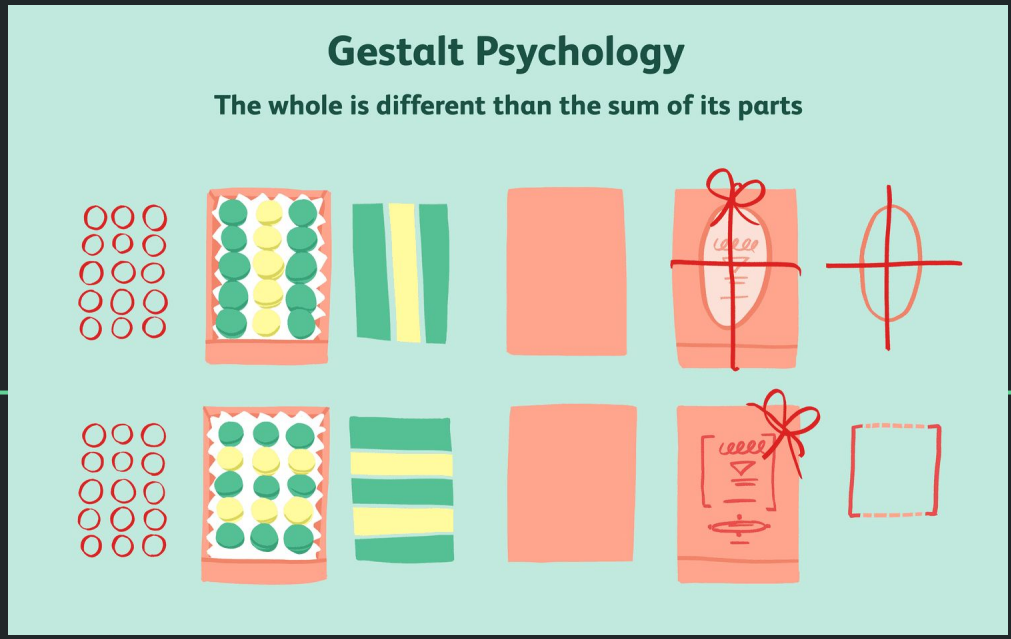
Gestalt:

- German word meaning "form,"

Refers to the tendency of the human mind to perceive the world in terms of meaningful patterns and wholes, rather than as a collection of isolated elements.

- We tend to group things together based on their shared characteristics or relationships, rather than treating them as individual entities.

- There is a school of psychology that deals with it.



Source: verywellmind.com

Gestalt

- has far-reaching implications for understanding how we process information and make decisions.

When receiving information from environment ...

... our brains don't just store and process individual pieces of information;

... our brains actively seek out patterns, relationships, and meaningful connections.

This ability to perceive the world in terms of Gestalts is essential for our survival and adaptation.

It allows us to quickly recognize objects and events, make sense of our surroundings, and learn from our experiences.

Universal Recoding

Information can be...

> ...transformed and recoded in various ways without losing its meaning.
>
> ...represented in different forms (numbers, symbols, images, or sounds), without affecting its essential content.

Example:

- A computer program can store information in binary code,
- Can also be represented in human-readable text or displayed as graphical elements.

➜ The meaning of the information remains the same, regardless of the form in which it is represented.

Universal Recoding

- Crucial for understanding how information is transmitted and processed in both biological and artificial systems.
- Allows for flexibility and adaptability
- Information can be transformed into different formats to suit the needs of the system.



Source: nature.com

Gestalt Principle: Organizing and Interpreting Information

- Provides the framework for structuring data
- Essential for coherent information processing

Universal Recoding: Flexible Transmission and Representation

- Enables adaptable communication and representation of information
- Fundamental for dynamic and responsive system behavior

- Together, these principles empower complex systems to process, store, and communicate effectively.
- They enhance adaptability, allowing systems to handle noisy data, adapt to changes, and learn from experiences.
- Fundamentally important in the realm of cybernetics and cybernetic systems.

Applications of Gestalt and Universal Recoding

The principles of Gestalt and Universal Recoding have broad applications in various fields, including:

Artificial Intelligence:

- develop AI systems that can perceive and understand the world in a more human-like way.

Machine Learning:

- train AI models to process and represent information in different ways,

Neuroscience:

- understand how the human brain processes information and organizes it into meaningful patterns.

Psychology:

- develop therapeutic techniques that help people process and cope with traumatic experiences.

- Wiener's work on Gestalt and universal recoding is foundational for understanding information processing, communication, and control.
- The insights derived from Wiener's principles have a broad impact on both natural and artificial systems.

- Wiener's principles provide valuable tools for the development of intelligent, adaptive, and human-centered technologies.
- The understanding of Gestalt and universal recoding continues to shape advancements in various fields, driving innovation and progress.

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VII - Cybernetics and Psychopathology

Norbert Wiener's Cybernetics

Chapter VII - Cybernetics and Psychopathology

Wiener discusses the parallels between the functionality of the brain and that of a computing machine

- Error Avoidance / Correction
- Malfunctions

Understanding the similarities between the brain and computing machines can offer valuable insights into psychopathology and psychiatry.

How does the brain avoid major errors and malfunctions that are caused by the failure of individual components ?

Error-Checking

Error-checking in computing machines:

- Checking every step of computation leads to increased complexity

Instead Wiener introduces a more efficient method

- 2 or 3 separate mechanisms do computations simultaneously
- The answers are compared
- Discrepancies allow the operators to locate and address the malfunctions

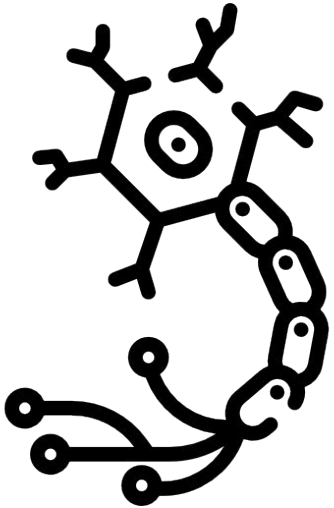
Error-Checking

Wiener draws parallels between Error-Checking in computing machines and potential mechanisms in the nervous system

The nervous system might incorporate...

... redundant elements

... alternative pathways



Internuncial Pools:

➜ Alternative routes for information transfer, especially in regions linked to higher mental functions

Functional and Organic Disorders

Wiener explores the relation between functional and organic disorders

- Functional disorders can be seen as diseases of memory
- Disorders impact the nervous system through disturbing the traffic rather than destroying tissue

Wiener explains that circular processes in a system with many neurons may be unstable for long periods

- The brain might lack the neurons for normal thought processes
- This leads to an overload that involves more neurons in the expanding process

Interrupting False Cycles

Computing Machines:

- Clear machine of all information
- Subject machine to impulses

Brain:

- Sleep
- Prefrontal Lobotomy
- Shock treatments (electric, insulin…)

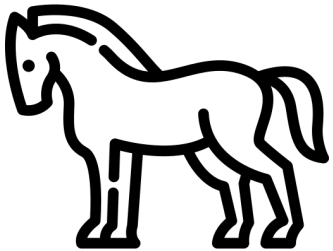


Source: dealnewsc.com

Wiener discusses the concept of traffic and overload in the nervous system

- Each form of organization has a limit beyond which it cannot function effectively

weight



Mechanism of trandering water



size



Wiener applies this idea to the nervous system and emphasizes the importance of the traffic problem

He compares the human nervous system to a telephone system:

- Limiting factor: Fraction of time a user can't make a call

- The more stages involved in a call, the more rapidly the service deteriorates



Overload

Wiener suggests that the human brain operates close to the edge of overload

Overload can occur due to:

1. Excessive traffic
2. Physical removal of channels
3. Occupation of channels by undesirable systems of traffic (extreme anxiety)

➜ When this happens there can be a sudden mental breakdown (possibly insanity)

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VIII - Language, Information, Society

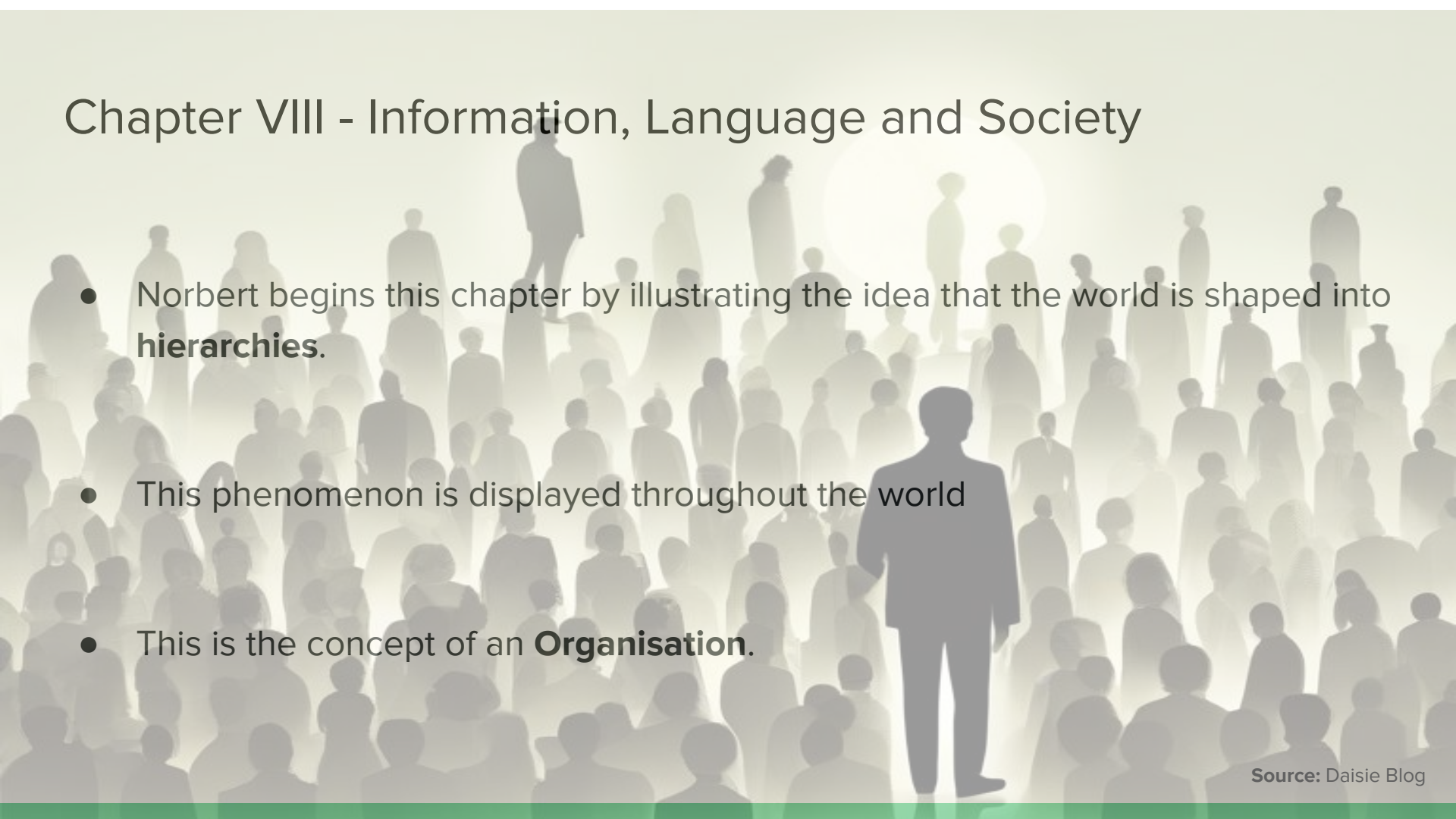
Norbert Wiener's Cybernetics

Chapter VIII - Information, Language and Society

- Norbert begins this chapter by illustrating the idea that the world is shaped into **hierarchies**.

- This phenomenon is displayed throughout the world

- This is the concept of an **Organisation**.



Source: Daisie Blog

- In illustrating this concept, Norbert draws attention to the beehive, and the **unity** it displays.
- **Intercommunication** between its members is key.
- Communication doesn’t necessitate language
- The exchange of **information** can be made in different ways .



- With humans, this communication mainly occurs through **language** and **literature**
- Humanity can be considered an organisation of humans
- Acquiring **information** also happens with **observation** and **emotions**.
- There are many **feedback** mechanisms defined in this organisation.

- Norbert relates the concepts of language, information and communication to **Society**.
- In order for a group to have a lot of information, its members need to be social so that the it can be spread between them.
- A group of non-social individuals may have very little information, but it is not necessarily the case for the individuals themselves.
- In the case of humans which are very social, there is an abundance of information present and available to their group.



- Norbert also illustrates from an outside perspective how **communities** work, and specifically small or close-knitted **communities**.
- These are more **homeostatic** than others, same whether the individuals forming them are literate or “savages”.
- These **communities** therefore work more in unity and are stable compared to large ones.
- Main ways of breaking unity, “anti-homeostatic factors”, controlling the means of **communication**.



Source: corporate.target.com

- Indeed, in larger societies, the press, books, telephone systems and the likes that take the role of spreading information.
- An organism can maintain homeostasis by having the means of acquiring, using, holding and transmissitting information
- This is the case of human **Society**, whose unity depends on these factors
- It is influenced by the actors of transmitting information within it.



Norbert appears to criticize the misuse of these means, by factors who seek to **control** and gain **power**.

Acquiring natural and human resources for themselves.

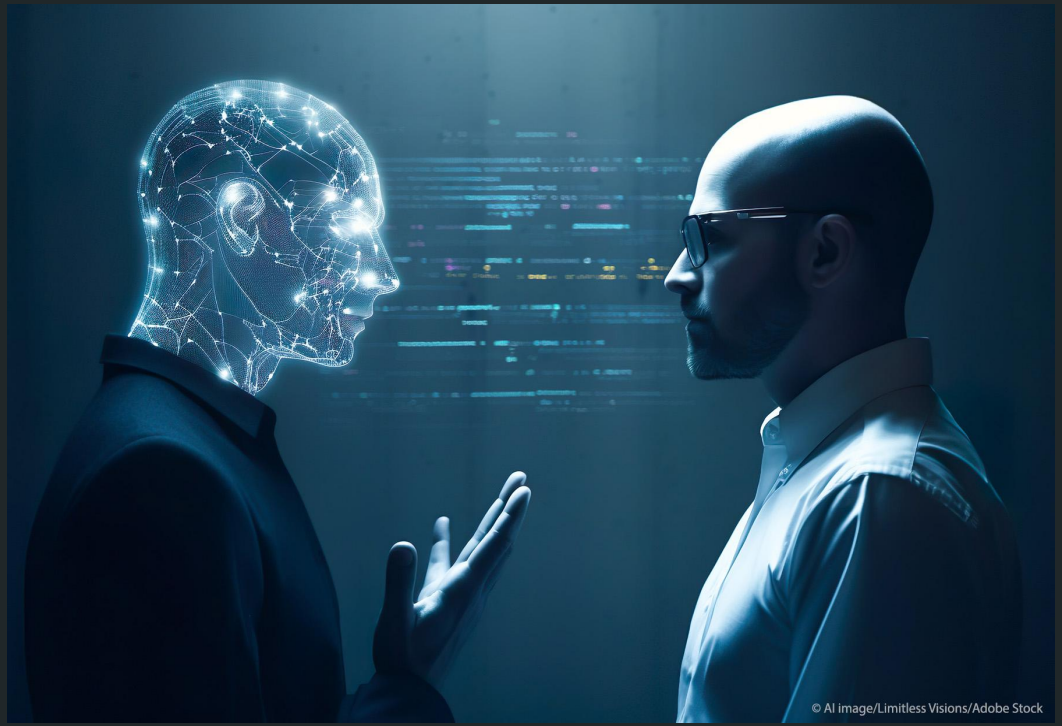
This can be related to the **demands** of these means of communication

In this case whoever can **influence** these factors is in control.

- Communication is one of the means of achieving homeostasis
- Misused by those seeking power and wealth and playing this game, thus harming the unity of Society, and breaking unity.
- This chapter deals with the ethical and societal implications of applying these cybernetic principles to social control.
- Concerns raised about the potential for automated systems and feedback loops to manage and regulate social behavior
- Potential erosion of individual autonomy and the suppression of dissent.

- Wiener's warnings continue to resonate in our increasingly digital and data-driven society. Especially with the emergence of AI.

- This chapter offers a thought-provoking examination of the intersection of information, language, and society.



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Smart Home Simulation

Cybernetics - Project



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1. Introduction

Smart Home Simulation - Project

1. Introduction

- Simulation of a home, equipped with various IoT devices (Smart Home)

Goals:

- Develop a realistic smart home environment
- Implement a diverse range of IoT devices
- Create an UI that allows to interact with the smart home
- Integrate Notification system to inform about changes, events or activities
- Allow users to create automations
- Integrate fundamental cybernetic principles

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2. Cybernetic Principles

Smart Home Simulation - Project

2. Cybernetic Principles

Feedback Loops:

- Smart Home adapts to user interactions / environmental changes

Automation and Control:

- Smart Home automates routine tasks
- Users exert control over the system

Adaptive Systems:

- Smart Devices and Automations adapt to changes

System Dynamics:

- Smart Devices are interconnected
- Devices form a system with dynamic behaviors

Information Processing:

- Smart Home processes information from various sensors

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3. Technologies Used

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3. Technologies used

Inkscape: GUI and Assets



Python: Code



Pygame: Framework



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4. Project Overview / Structure

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4. Project Overview / Structure

1. Smart Home
2. Smart Devices
3. Smart Home App
4. Automations

Smart Home

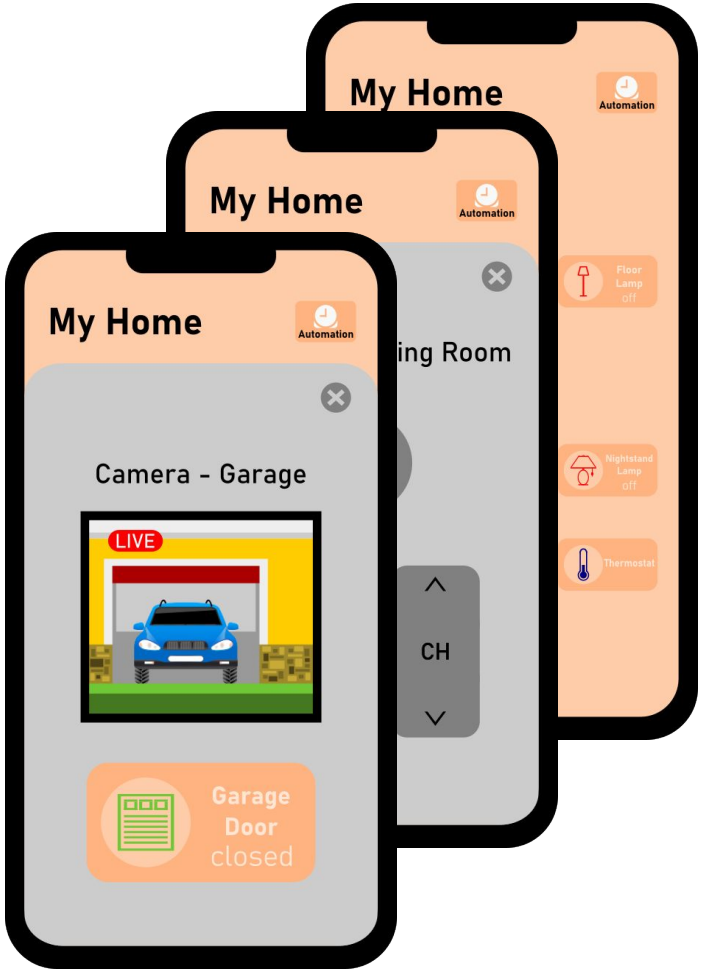
- The Smart Home is displayed in frontal view.
- This allows the depiction of each room of the household, as well as the outside premises.
- The interactions and the ensuing changes are shown directly in this representation.



The Smart Home App

The Smart Home is controlled through the app:

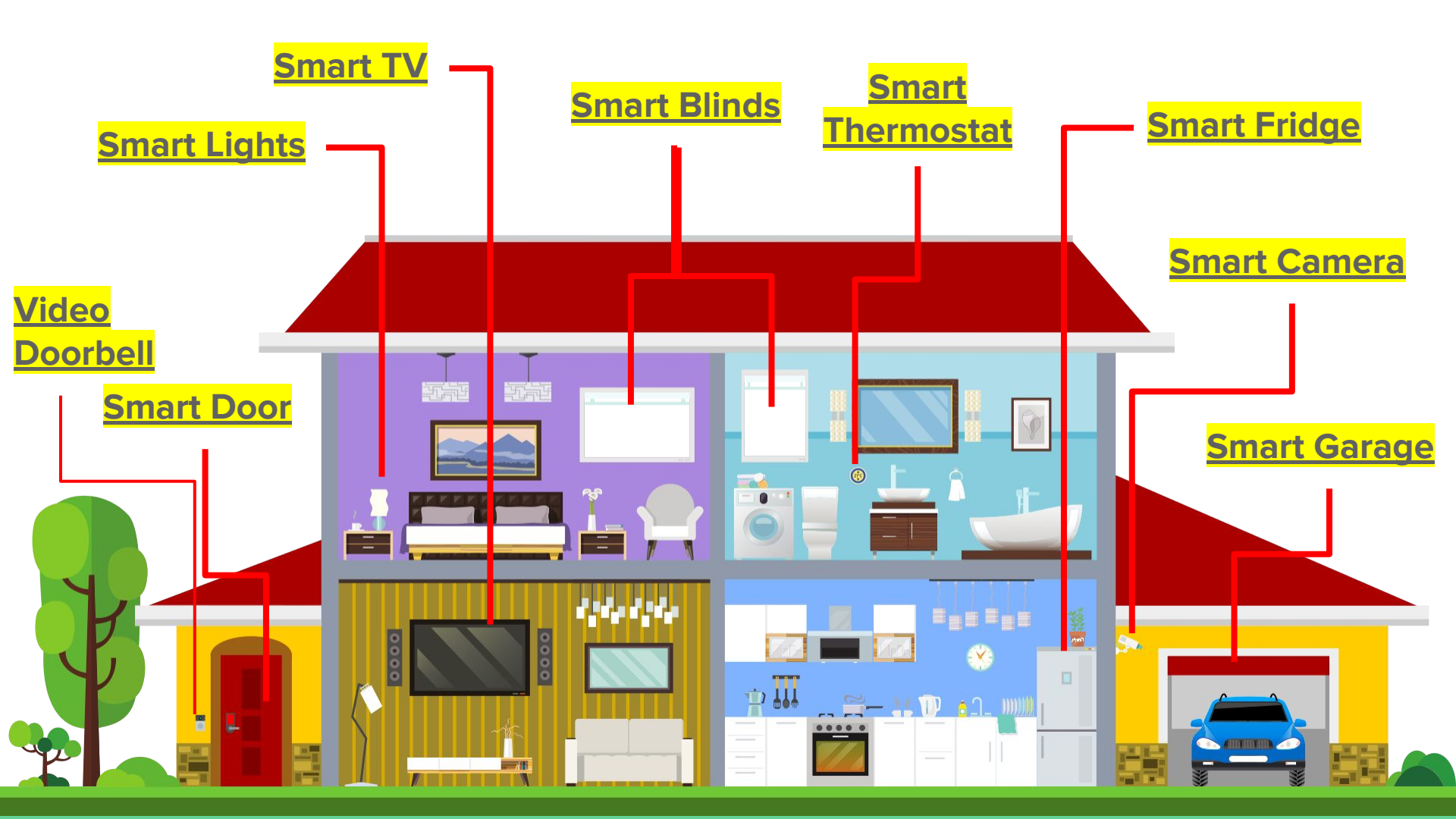
- Modify variables and receive information
- Control smart devices/aspects of the household
- Receiving messages and notifications



Devices

- The main feature of the Smart Horme are the smart devices.
- They are the typical household furniture/appliances, but that come with smart features that can be controlled in the app.
- Every room has at least one of these implemented, as well as the outside doors.



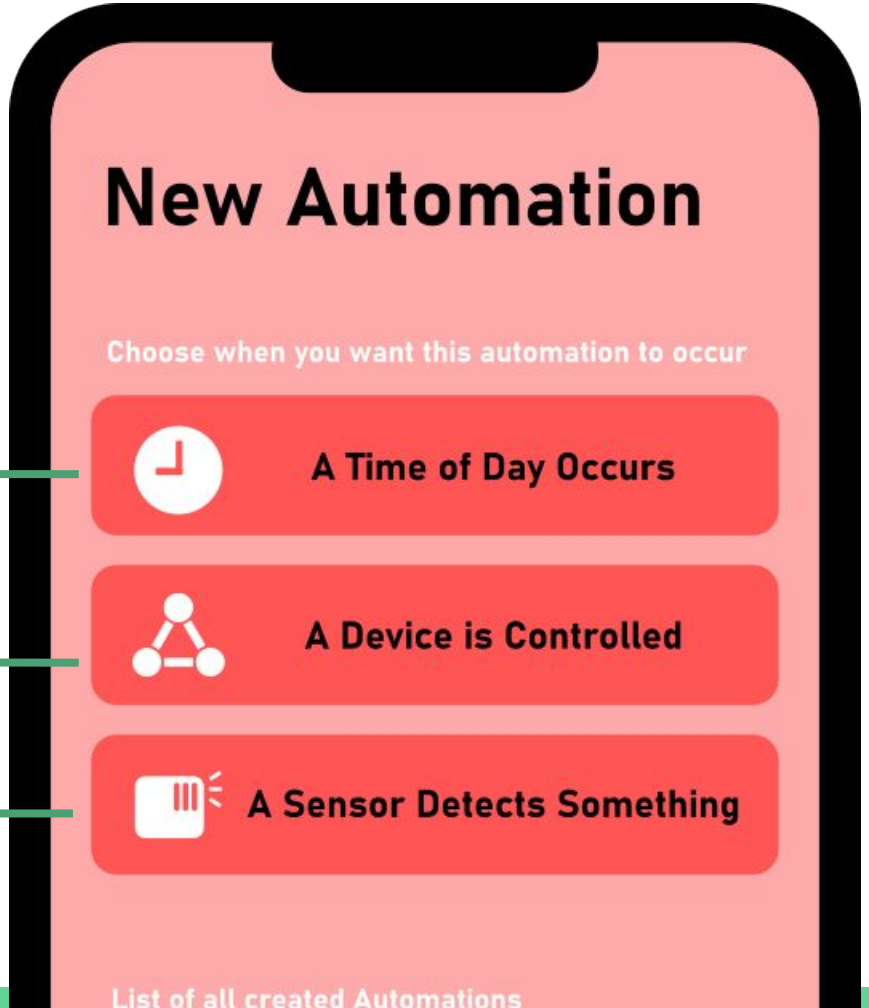

Smart TV
Smart Lights
Smart Blinds
Smart Thermostat
Smart Fridge
Smart Camera
Video Doorbell
Smart Door
Smart Garage

Automations

Automatically initiate actions based on specific times of day

Establish interactions and responses between different smart devices

Trigger actions based on input from various sensors detecting changes



- Through the app the user may make a list of settings that he/she wants to be implemented.
- The implementation can be set based on time, a device being controlled or sensor detection/feedback
- All automations can be viewed in the app



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5. Demo

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The End! - Any Questions?

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